

Maven Fuzzy Factory

Business Performance Analysis

Data Analytics Report

Excel • Power BI

Portfolio Project

Bao Khue Cao

July 2026

1. Executive summary

This project analyzes the performance of Maven Fuzzy Factory using sales, marketing, website, and refund data. The objective is to identify the key drivers of revenue, evaluate marketing effectiveness, and uncover operational issues affecting profitability.

Exploratory analysis revealed that revenue is highly concentrated in the flagship product (Mr. Fuzzy), website traffic and conversion rates improved steadily over time, Google Search remained the dominant acquisition channel while Bing demonstrated higher conversion efficiency, and refund performance remained generally stable despite temporary spikes.

In order to communicate these insights effectively, an Excel dashboard was developed to summarize product performance and key financial metrics using Pivot Tables and charts, while an interactive Power BI dashboard was created to monitor sales trends, marketing performance, customer behavior, and refund metrics through dynamic filtering and drill-down capabilities.

2. Business Problem

Maven Fuzzy Factory sells multiple toy products through online marketing channels. This analysis aims to answer the following business questions:

- Which products generate the greatest business value?
- Which marketing channels are most effective?
- Where are refund-related revenue losses occurring?

3. Dataset Overview

E-Commerce database for Maven Fuzzy Factory, an online retailer that sells teddy bears. It includes detailed marketing data on website sessions and pageviews by user, as well as their orders and returns.

The dataset was provided by Maven Analytics: [Maven Analytics](#)

Table	Description
<i>orders</i>	Order details
<i>order_items</i>	Items purchased
<i>order_item_refunds</i>	Refund items details
<i>products</i>	Product information
<i>website_sessions</i>	Sessions across marketing channels
<i>website_pageviews</i>	Pageviews details

4. Data preprocessing

- NULL values were replaced with 'Unknown'
- Fixing incorrect data types
- Add calculated columns with business metrics
- Add date hierarchy table
- Add measures to calculate business metrics

5. Exploratory Analysis

5.1. Sales Overview Analysis

Key Metrics

Metric	Value
Total Orders	32K
Net Revenue	\$1.85M
Net Profit Margin	61.02%

Findings

1. Strong long-term business growth

Both net revenue and orders show a clear upward trend throughout the period. Revenue increased from only a few thousand dollars per month in early 2012 to over \$130K at its peak. Monthly orders followed the same pattern, growing from fewer than 100 orders to over 2,000 orders.

2. Revenue growth closely follows order growth

The revenue trend and order trend move almost identically. Periods with higher order volumes consistently generate higher revenue.

3. Profitability remains consistently high

Despite fluctuations in revenue, the overall net profit margin remains around 61%. The business maintains healthy profitability while scaling sales.

4. Evidence of seasonality

Revenue and orders both peak during late 2014 and early 2015 before declining afterward. The decline occurs simultaneously across both metrics.

Business implication:

The business experienced sustained expansion, indicating successful customer acquisition and increasing demand. It can be seen that the close relationship between revenue and order growth suggests that business expansion has primarily been driven by increasing sales volume rather than higher prices. Growth has not come at the expense of profitability, suggesting effective pricing

and cost control. Besides, the spike in revenue and orders likely reflects seasonal demand or promotional campaigns rather than a permanent increase in sales.

Recommendations

- Investigate the causes of the late-2014 sales surge and determine whether similar campaigns or seasonal strategies can be replicated.
- Monitor whether future revenue growth continues to be driven primarily by higher order volume or whether average order value begins to change.
- Maintain current pricing and cost management practices to preserve the strong profit margin while expanding sales.

5.2. Product Performance Analysis

Key metrics

Metric	Value
Total Items Sold	40.0K
Total Gross Revenue	\$1.9M
Total Gross Profit	\$1.2M
Gross Profit Margin	63%

Findings

1. Revenue is highly concentrated in a single product

Mr. Fuzzy generated approximately 62% of total revenue and sold nearly 25,000 units, significantly outperforming all other products.

2. Sales volume and revenue show a similar pattern

Mr. Fuzzy recorded the highest sales volume and revenue, while Love Bear, Sugar Panda, and Mini Bear contributed relatively smaller shares.

3. High-selling products are not necessarily the most profitable

Although Mr. Fuzzy generated the highest profit in absolute terms, it did not achieve the highest profit margin. Products such as Sugar Panda and Mini Bear achieved profit margins close to 69%, compared to approximately 61% for Mr. Fuzzy.

4. Profitability remains strong across the product portfolio

All products achieved gross profit margins above 60%, indicating a generally healthy product mix and pricing strategy.

Business implication:

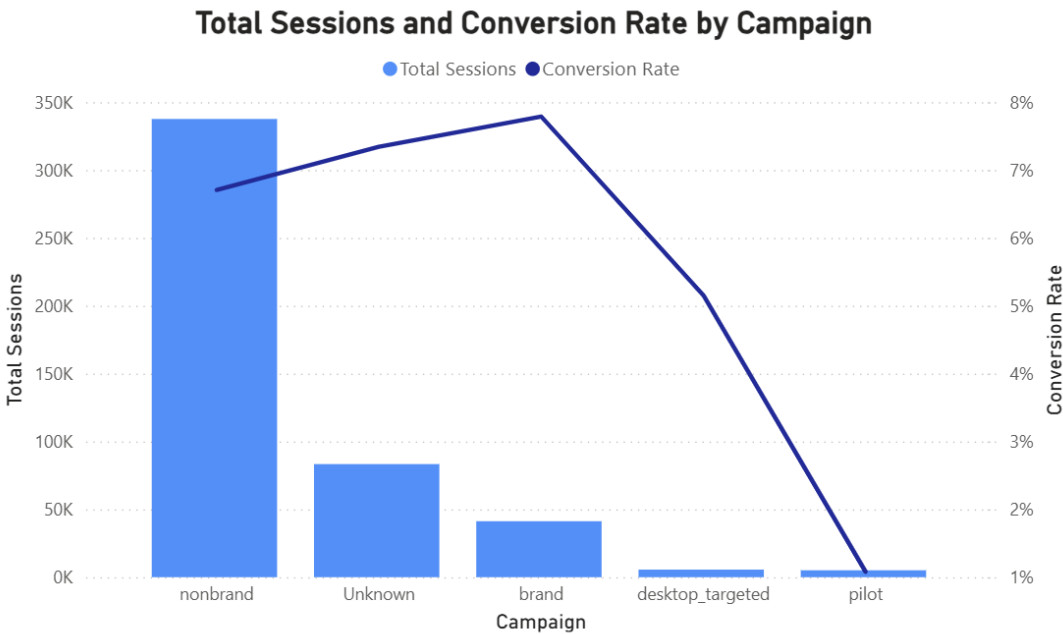
Revenue is highly concentrated in Mr. Fuzzy, creating concentration risk. Any decline in demand, stock availability, or customer satisfaction for this product would have a disproportionate impact on company performance. Therefore, future revenue growth may require either increasing sales volume of secondary products or reducing reliance on the top-selling product. Furthermore, some lower-volume products generate greater profit per dollar of revenue and may deserve additional marketing attention.

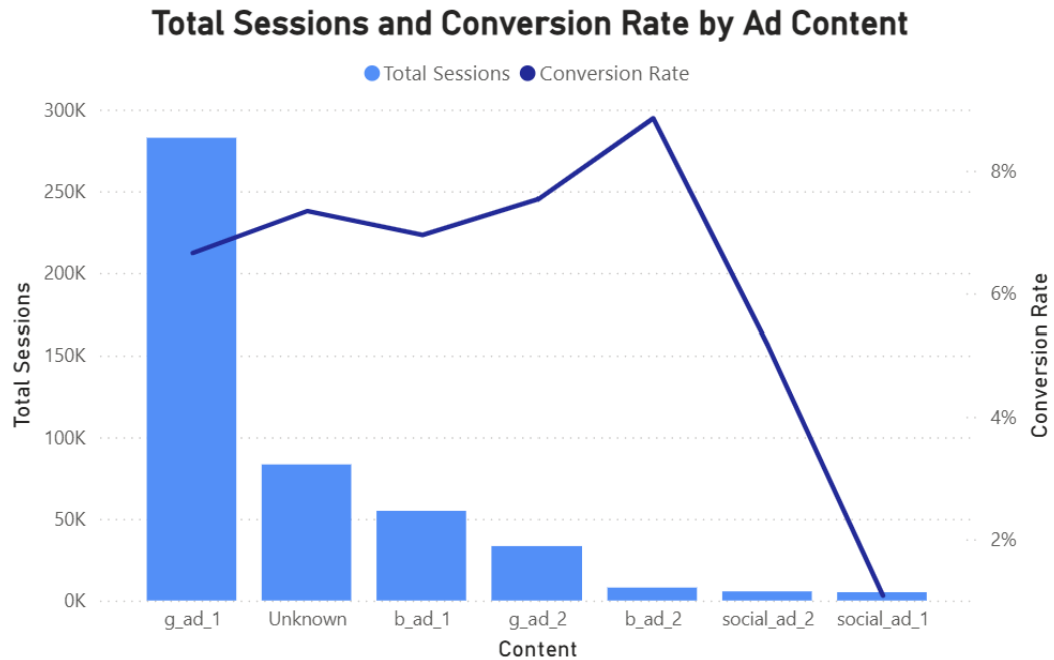
Recommendations

- Continue supporting Mr. Fuzzy as the flagship product due to its dominant contribution to revenue and profit.
- Investigate opportunities to increase visibility and sales of Sugar Panda and Mini Bear, which exhibit strong profit margins.
- Monitor revenue concentration risk and diversify revenue sources across the product portfolio.
- Evaluate whether higher-margin products can be promoted more aggressively to improve overall profitability.

5.3. Marketing & Website Behavior

5.3.1. Marketing analysis





Findings

1. Traffic Growth

Monthly traffic increased nearly tenfold over the observed period. There were two major traffic spikes in Q4 of 2012 and 2014.

2. Conversion Trend

Conversion rate improved from about 3% to over 8%, improvement was gradual and sustained.

3. Channel Performance

Google Search generated the highest number of sessions, approximately 316k and the most revenue, roughly \$1.2M. Bing Search generated fewer sessions, at about 62k, but a higher conversion rate of 7.2% compared to 6.75% of Google Search. Moreover, sessions without a tagged traffic source ("Unknown") accounted for approximately 17.6% of total traffic and recorded the highest conversion rate of any group, at 7.34%. Social traffic had both low traffic volume and low conversion.

4. Campaign Performance

Nonbrand campaigns generated the largest traffic volume, around 337k. Brand campaigns achieved the highest conversion rate of 7.79%. Desktop-targeted and pilot campaigns had relatively weak performance.

5. Device Performance

Desktop accounts for most sessions, approximately 70%. Desktop conversion rate is substantially higher than mobile, at 8.5% compared to 3.1% of mobile.

6. Ad Content Performance

Different ad creatives show clear differences in conversion. Some creatives attract high traffic but convert less efficiently, while g_ad_1 have a total session of 282k, its conversion rate is only 6.66%. On the other hand, b_ad_2 has only 7k9 total sessions but a conversion rate of 8.86%.

Business implication:

From the analysis, it is clear that marketing efforts successfully increased website traffic over time. Looking at the increase in conversion rate, the company has become more effective at turning visitors into customers.

Regarding channel performance, Google is the primary acquisition channel. Bing may represent an efficient acquisition channel and warrants further investigation before reallocating marketing budget. Social campaigns should be reviewed before increasing the budget. Additionally, the substantial share of untagged ("Unknown") traffic, which also converts unusually well, suggests a possible tracking or attribution gap worth investigating.

Regarding campaign performance, nonbrand is effective for acquisition while brand campaigns are effective for converting users already familiar with the company.

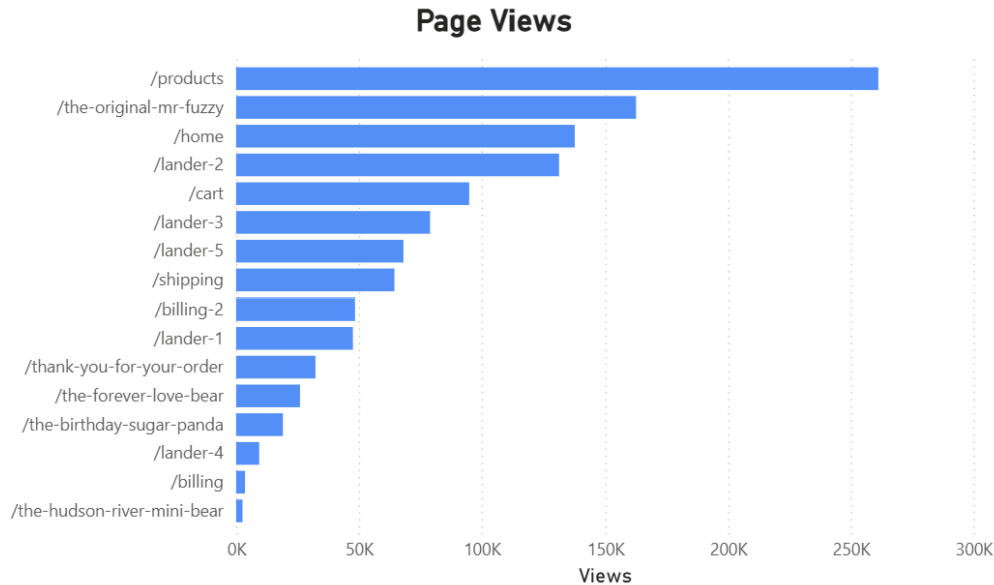
Lastly, about device performance, mobile experience may need optimization. Desktop currently delivers most of the business value.

In conclusion, marketing should evaluate campaigns on both **volume** and **efficiency**, not just sessions.

Recommendations

- Continue investing in Google Search due to its dominant traffic and revenue contribution.
- Test increasing Bing advertising budget given its relatively high conversion efficiency.
- Investigate why desktop converts significantly better than mobile and optimize the mobile experience.
- Investigate the source of "Unknown" (untagged) traffic.
- Review low-performing social campaigns before allocating additional marketing expenditure.
- Analyze successful ad creatives to identify characteristics that can be applied to future campaigns.

5.3.2. Website behavior analysis



Findings

1. Bounce rate is moderate

The website recorded a 44.76% bounce rate, meaning nearly half of all sessions ended after viewing only one page.

2. Visitors primarily enter through product-related pages

The /products page received the highest number of page views, at 261k views, followed by: /the-original-mr-fuzzy, /home, and several landing pages.

3. The product interest matches sales performance

The Mr. Fuzzy product page received substantially more views than the pages for Love Bear or Sugar Panda. This aligns with the earlier product analysis showing that Mr. Fuzzy generates the highest sales and revenue, suggesting customer attention is translating into purchases.

4. Landing pages contribute meaningful traffic

Several landing pages (/lander-2, /lander-3, /lander-5) rank among the most visited pages.

5. Purchase funnel pages receive progressively fewer visits

Pages such as Cart, Shipping, and Billing receive considerably fewer page views than product pages.

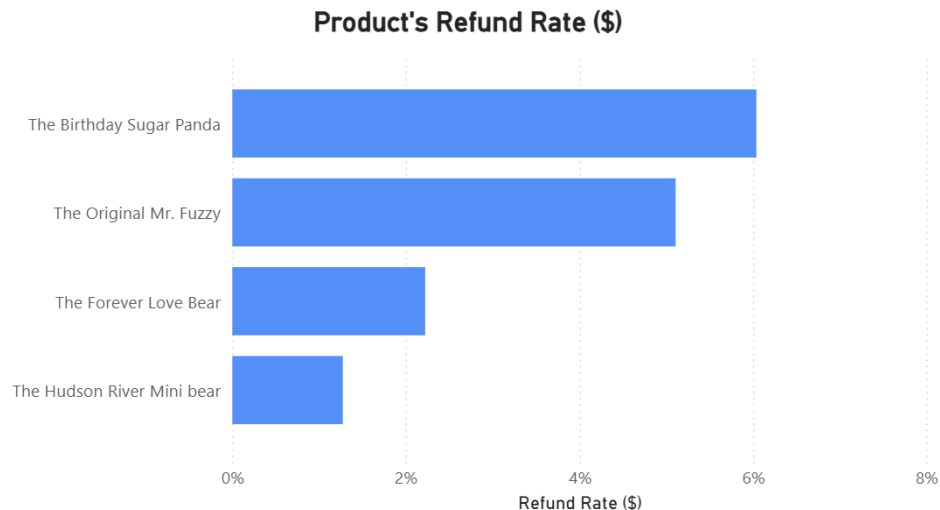
Business implication:

Visitors are generally willing to explore the website, but there is still room to improve engagement and encourage users to continue browsing. From the findings, it is clear that product pages play a key role in attracting visitors and should remain a focus for optimization. Additionally, traffic distribution appears consistent with sales performance, indicating that marketing efforts are successfully directing customers toward the flagship product. Moreover, dedicated landing pages are successfully driving campaign traffic and should continue to be monitored for conversion performance.

Recommendations

- Continue optimizing high-traffic product pages, particularly Mr. Fuzzy.
- Improve landing page quality to increase visitor engagement and downstream conversions.
- Investigate opportunities to reduce bounce rate through stronger calls-to-action, internal linking, or personalized recommendations.

5.4. Refund analysis



Findings:

1. Overall refund performance

Overall refund amount is \$85.34K, representing a 4.40% refund rate. While refunds are not excessively high overall, there are periods where refund performance deteriorated significantly.

2. Refund Trend

Refund rate is highly volatile, ranging from roughly 2% to over 9% across the observed period. Two major spikes occurred in Q3 2012, where the refund rate approached 9% and in Q3 2014, the refund rate again exceeded 9%.

3. Product Performance

The Original Mr. Fuzzy has the highest refund rate (around 3%), substantially higher than the other products. Despite being the top-selling product, it also contributes the largest share of refunds.

Business implication:

Although refunds account for a relatively small proportion of revenue, periods of elevated refund activity indicate that refund performance should continue to be monitored. The absence of sustained high refund rates suggests these spikes were temporary events rather than a persistent operational issue. However, the business should investigate similar future spikes promptly to

minimize revenue loss. Because the company's revenue is heavily dependent on The Original Mr. Fuzzy, reducing refunds for this product could improve both profitability and customer satisfaction.

Recommendations:

- Investigate the periods with unusually high refund rates (Q3 2012 and Q3 2014) to determine whether they were caused by operational, fulfillment, or product-related issues.
- Prioritize monitoring of The Original Mr. Fuzzy, as reducing refunds for the highest-selling product would have the greatest financial impact.
- Track refund rate as a recurring KPI to identify abnormal increases early.

6. Conclusion

The analysis demonstrates that Maven Fuzzy Factory achieved sustained growth in revenue, traffic, and conversion while maintaining strong profitability. However, revenue concentration in Mr. Fuzzy, lower mobile conversion performance, and temporary refund spikes represent opportunities for further investigation and optimization.

The findings informed the development of two complementary dashboards. An Excel dashboard provides a concise management summary of product performance and financial KPIs, while the Power BI dashboard offers an interactive environment for monitoring sales, marketing effectiveness, customer behavior, and refund performance. Together, these dashboards support data-driven decision-making by combining quick executive reporting with deeper analytical exploration.